

Introduction to Deep Learning

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3/13/2024



Agenda

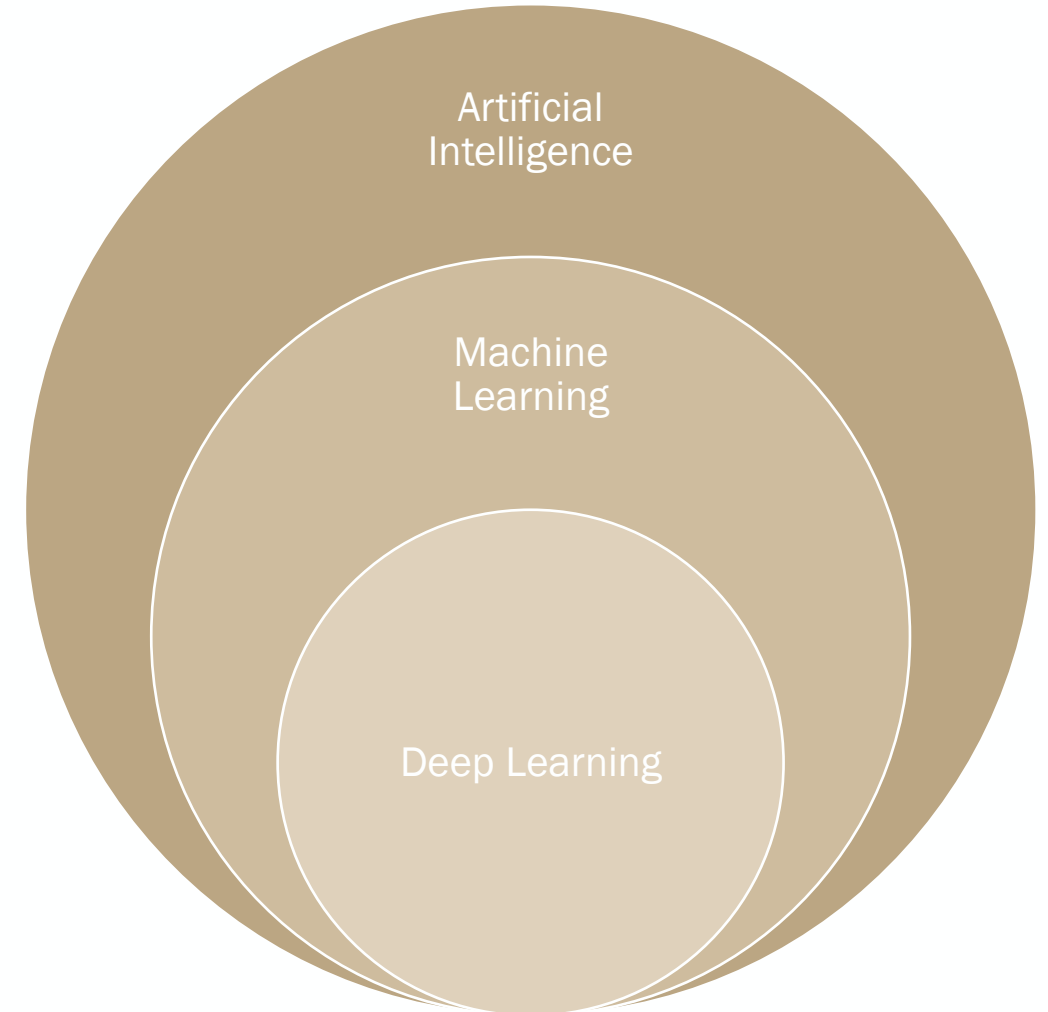
- What is Deep Learning?
- The Multi-Layer Perceptron
- Loss Functions
- Training Neural Networks
- Convolutional Neural Networks and Sequence Models
- Code Demo

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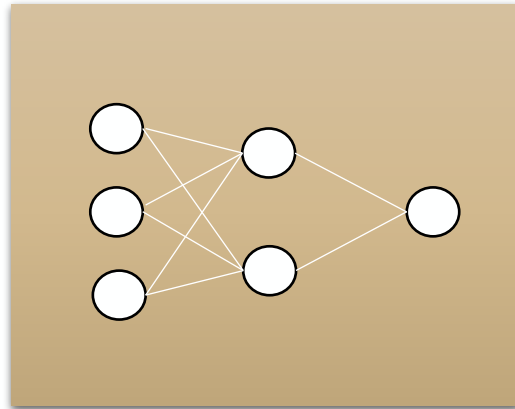
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What is Deep Learning?

- Artificial intelligence refers to methods that enable machines to perform tasks that require human intelligence.
- Machine learning refers to algorithms that allow machines to learn from data or experience without being explicitly programmed.
- Deep learning refers to utilizing multi-layered artificial neural networks to learn patterns from data.



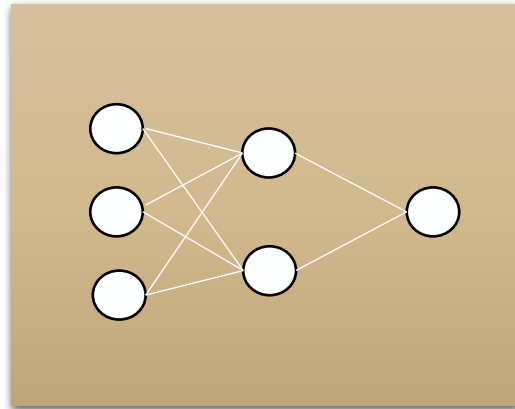
Deep Learning Applications



Cat 98.2%
Dog 1.8%

Deep Learning Applications

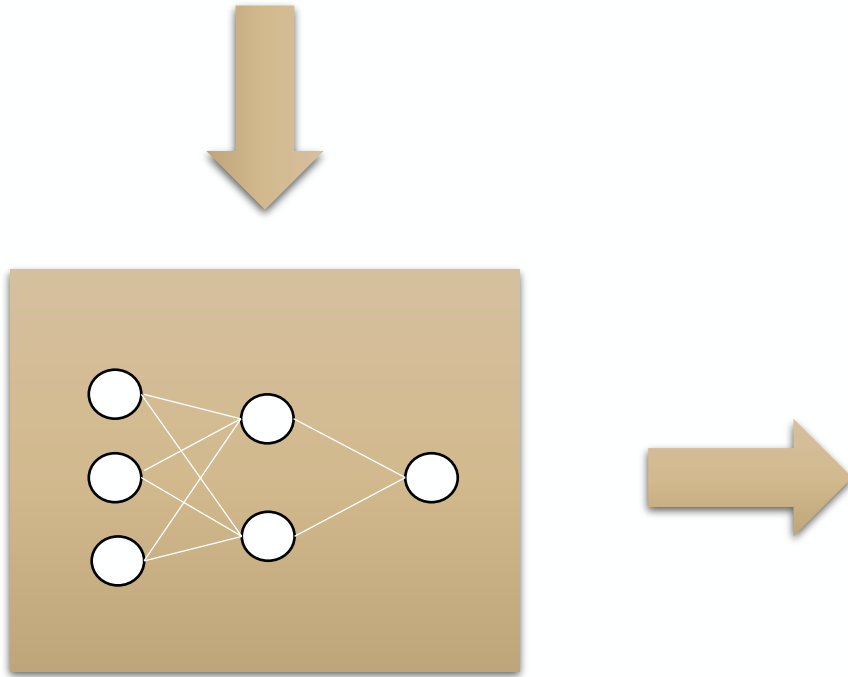
Good morning



Bonjour

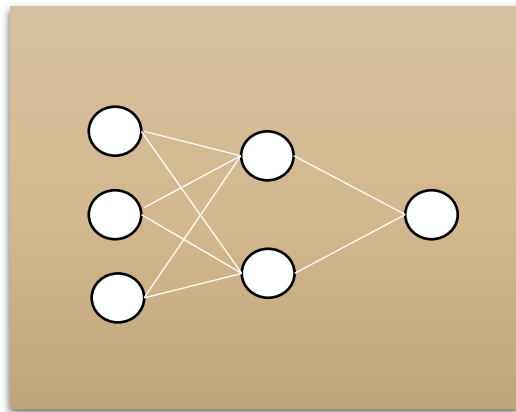
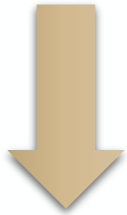
Deep Learning Applications

A panda riding a bike



Deep Learning Applications

List some deep learning applications



ChatGPT

Certainly! Deep learning has found applications across various domains. Here are some examples:

1. Computer Vision:

- Object Detection: Identifying and localizing objects within images or videos.
- Image Classification: Assigning labels to images based on their content.
- Facial Recognition: Identifying and verifying individuals' faces.
- Image Segmentation: Dividing an image into semantically meaningful parts.
- Autonomous Vehicles: Deep learning is crucial for perception tasks in self-driving cars.

2. Natural Language Processing (NLP):

- Machine Translation: Translating text from one language to another.
- Sentiment Analysis: Determining the sentiment or opinion expressed in text.
- Named Entity Recognition (NER): Identifying and classifying entities mentioned in text (e.g., people, organizations, locations).
- Chatbots: Creating conversational agents that can interact with users in natural language.
- Text Summarization: Generating concise summaries from longer texts.

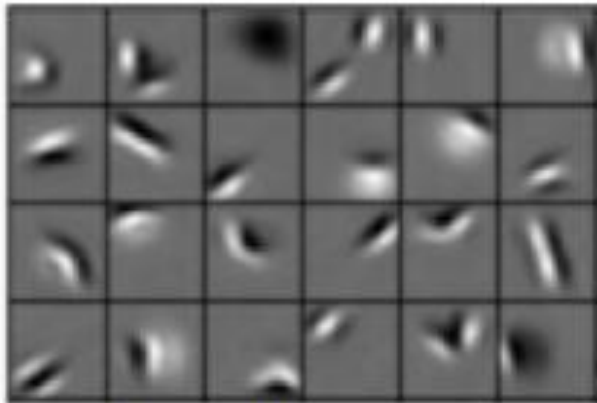
3. Speech Recognition and Synthesis:

- Speech-to-Text: Converting spoken language into text.
- Text-to-Speech: Generating human-like speech from text input.
- Speaker Recognition: Identifying or verifying the identity of a speaker from their voice.

Why Deep Learning?

- Deep learning removes the need for handcrafted features.

Low level features



Edges, dark spots

Mid level features



Eyes, ears, nose

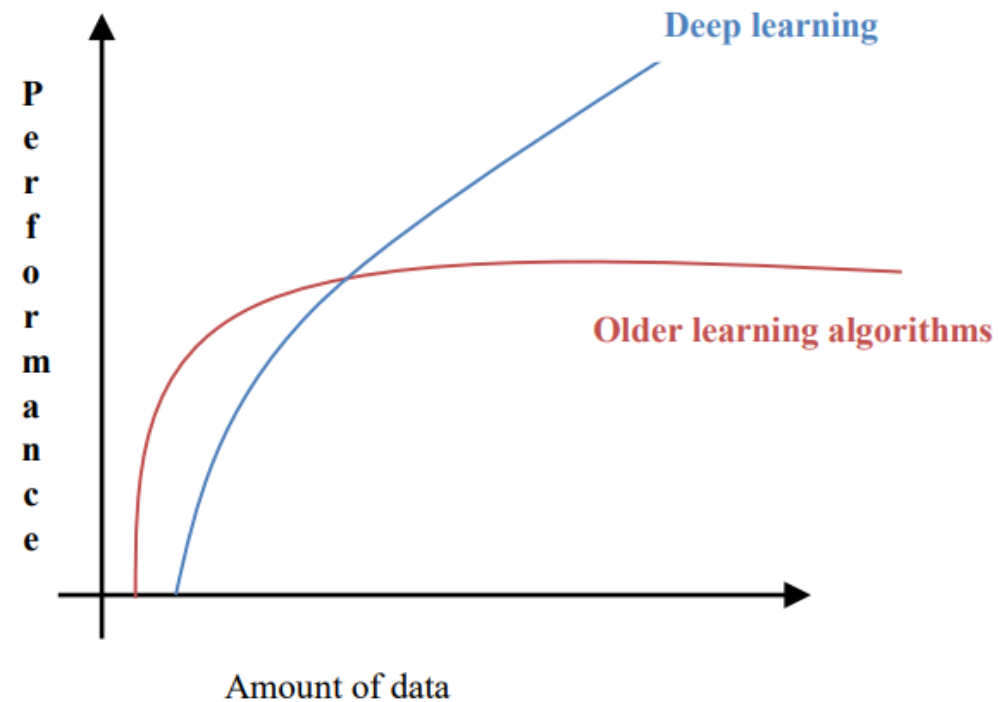
High level features



Facial structure

Why Deep Learning?

- Deep learning scales better with more data.

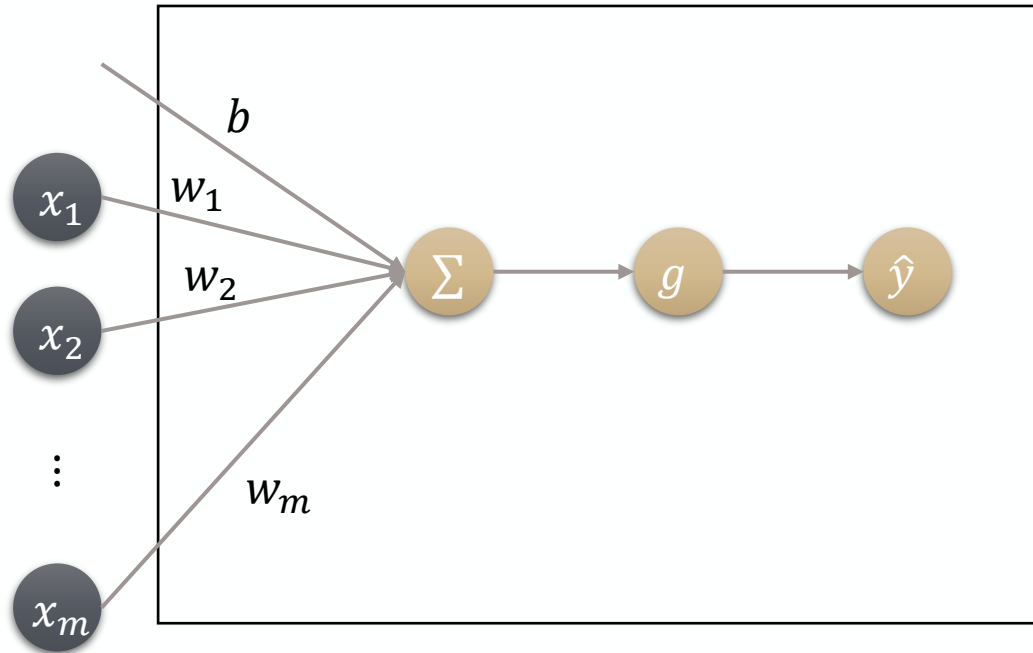


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The Perceptron

The building block of neural networks

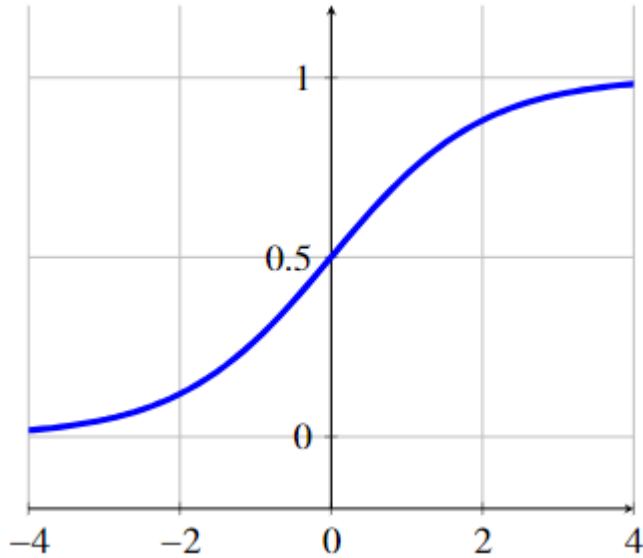


$$\hat{y} = g \left(\sum_{i=1}^m w_i x_i + b \right)$$

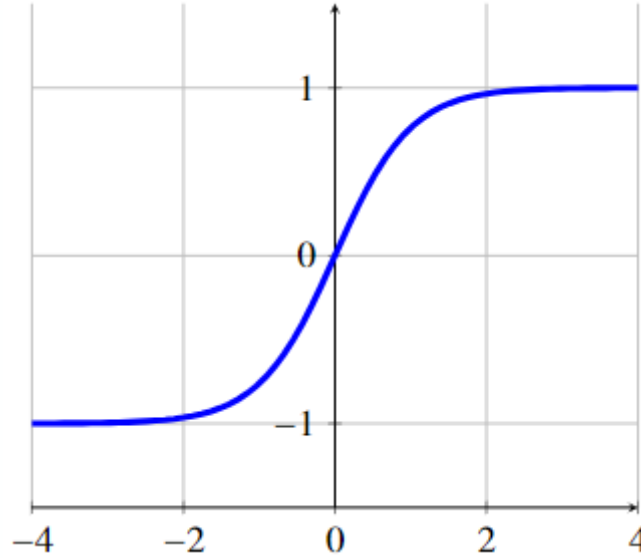
$$\hat{y} = g(X^T W + b)$$

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}, W = \begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix}$$

Activation Functions



Sigmoid: $g(x) = \frac{1}{1+e^{-x}}$

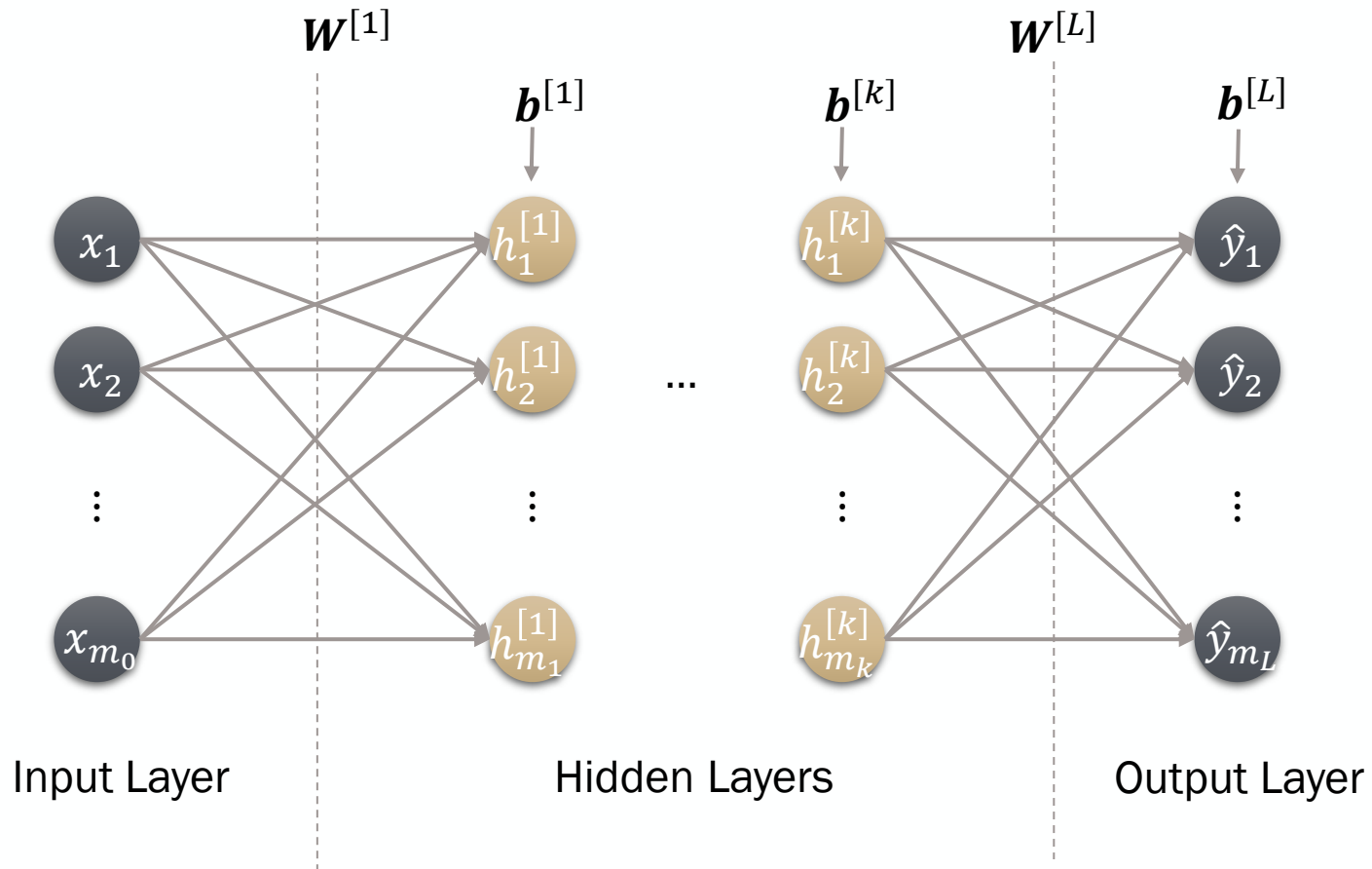


Tanh: $g(x) = \tanh(x)$



Relu: $g(x) = \max(0, x)$

The Multi-Layer Perceptron



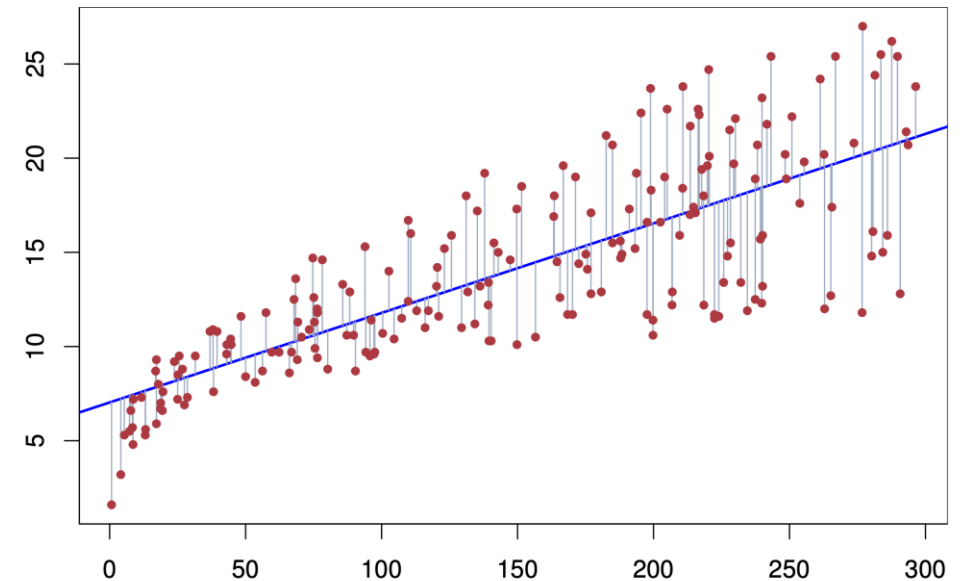
$$\begin{aligned} \mathbf{h}^{[0]} &= \mathbf{X} \\ \mathbf{h}^{[L]} &= \hat{\mathbf{Y}} \\ k &= L - 1 \\ \mathbf{h}^{[l]} &= g(\mathbf{W}^{[l]} \mathbf{h}^{[l-1]} + \mathbf{b}^{[l]}) \end{aligned}$$

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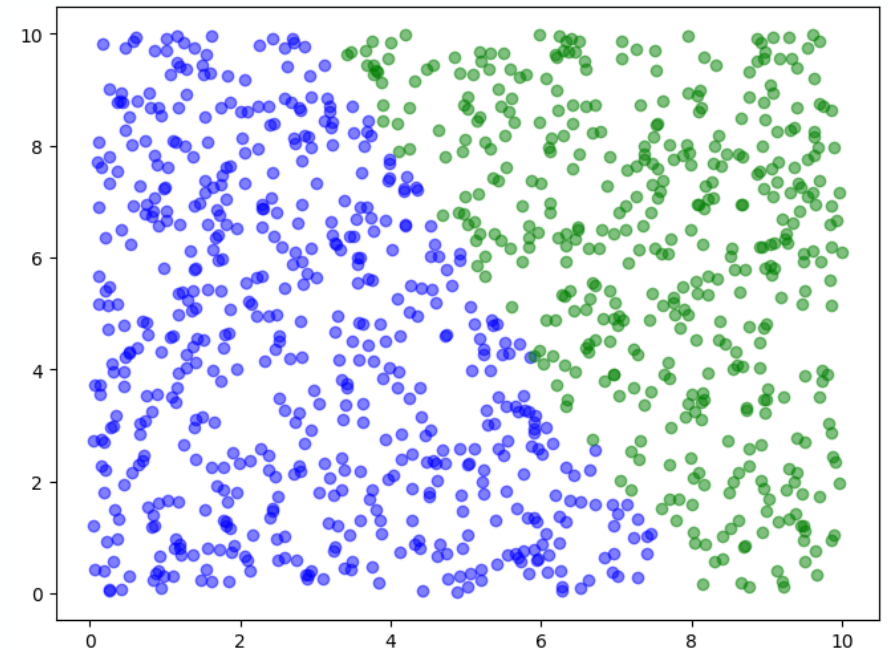
Mean Squared Error (MSE) Loss Function

- Given an input output pair (x_i, y_i) where x_i is a vector representing the features of the input and y_i is a real number representing the expected output, how to quantify the performance of a neural network?
- The mean squared error loss function can be used to evaluate and train the neural networks.
- $\mathcal{L}(f(x_i), y_i) = (y_i - f(x_i))^2$



Binary Cross Entropy Loss Function

- Given an input output pair (x_i, y_i) where x_i is a vector representing the features of the input and y_i is a binary value representing the point class, how to quantify the performance of a neural network?
- The binary cross entropy loss function can be used to evaluate and train the neural networks.
- $\mathcal{L}(f(x_i), y_i) = -\left(y_i \log(f(x_i)) + (1 - y_i) \log(1 - f(x_i))\right)$,



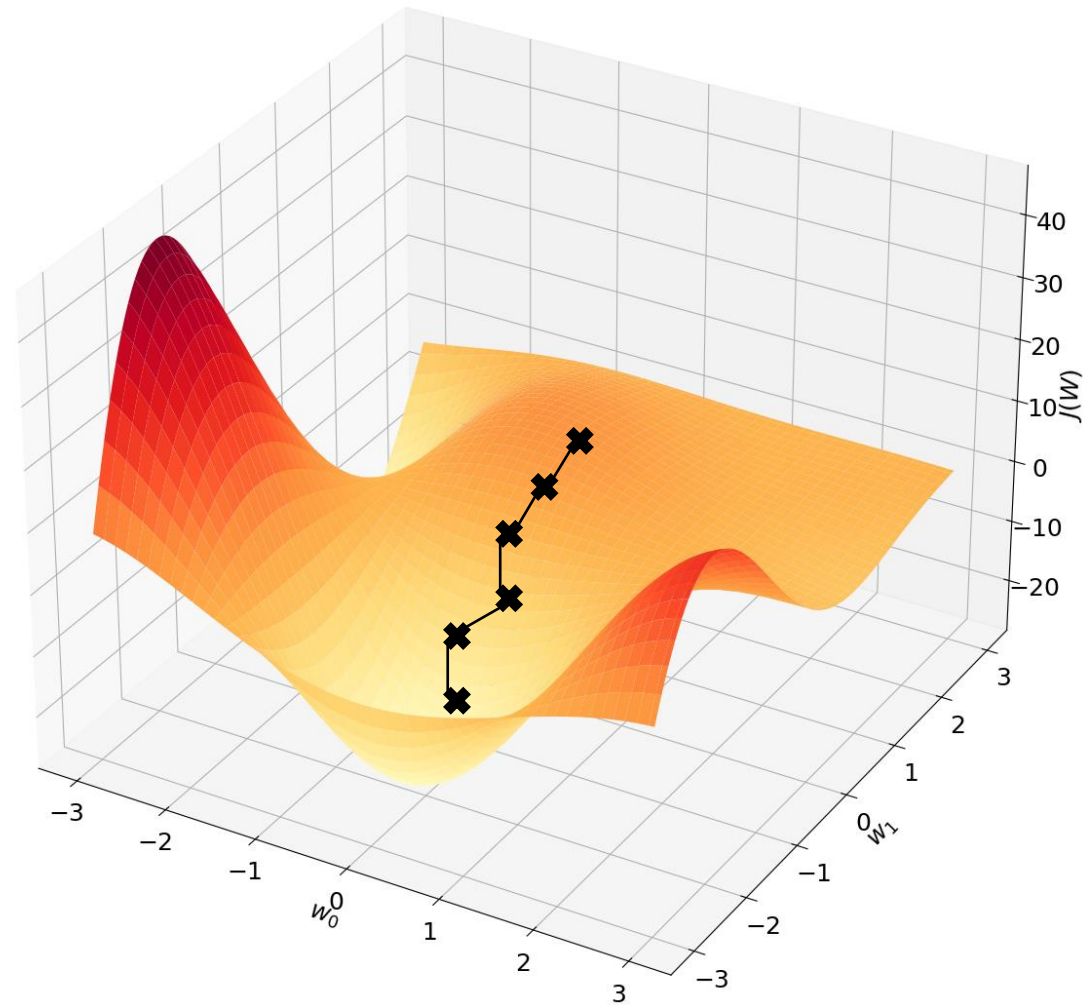
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Training Neural Networks

- The goal of training a neural network is to find the optimal weights minimizing the average loss across a dataset with n samples.
- $\mathbf{W}^* = \arg \min_{\mathbf{W}} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(f(x_i), y_i) = \arg \min_{\mathbf{W}} \mathcal{J}(\mathbf{W})$

Gradient Descent

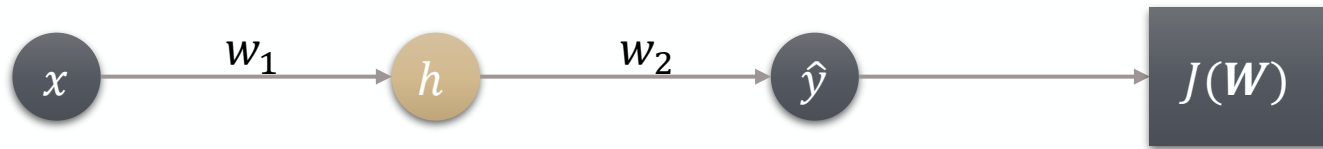


Gradient Descent

Algorithm

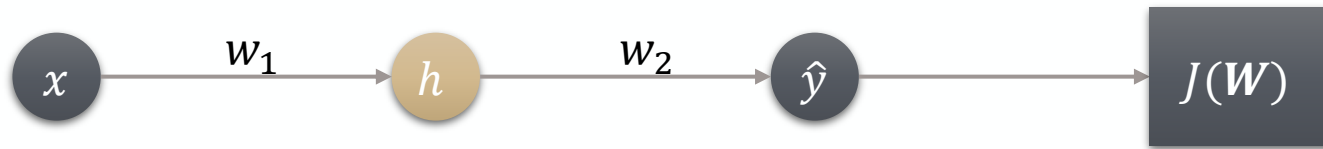
- Initialize weights randomly
- Repeat until convergence
 - $\mathbf{W} \leftarrow \mathbf{W} - \eta \frac{\partial J(\mathbf{W})}{\partial \mathbf{W}}$

Backward Propagation



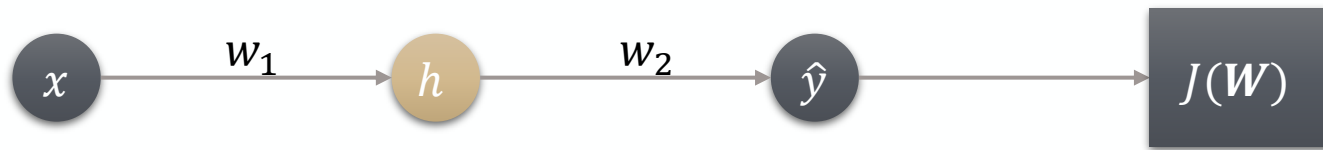
$$\frac{\partial J(W)}{\partial w_2} = \frac{\partial J(W)}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial w_2}$$

Backward Propagation



$$\frac{\partial J(W)}{\partial w_1} = \frac{\partial J(W)}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial w_1}$$

Backward Propagation



$$\frac{\partial J(W)}{\partial w_1} = \frac{\partial J(W)}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial h} * \frac{\partial h}{\partial w_1}$$

Stochastic Gradient Descent

- Computing the loss for the entire dataset in for the gradient descent algorithm might be computationally infeasible.
- Instead, stochastic gradient descent updates the parameters using only one sample

Algorithm

- Initialize weights randomly
- Repeat until convergence
 - Pick a random sample i
 - $W \leftarrow W - \eta \frac{\partial J_i(W)}{\partial W}$

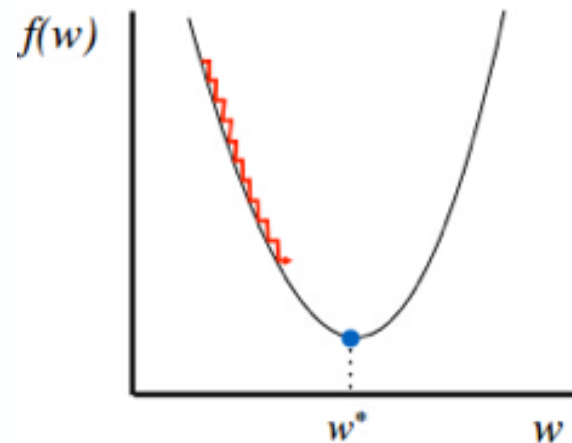
Batched Stochastic Gradient Descent

Algorithm

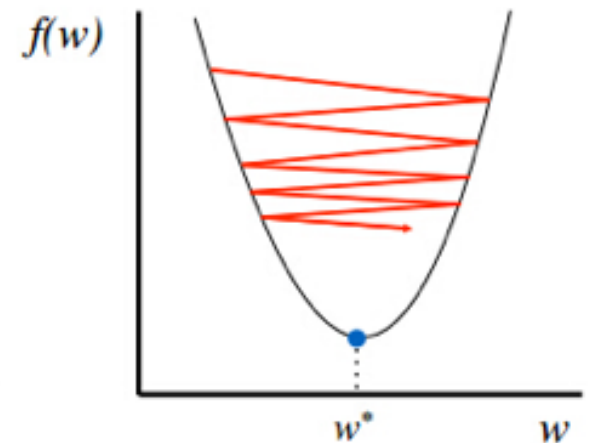
- Initialize weights randomly
- Repeat until convergence
 - Pick a random batch of B samples
 - $\mathbf{W} \leftarrow \mathbf{W} - \eta \frac{1}{B} \sum_{i=1}^B \frac{\partial J_i(\mathbf{W})}{\partial \mathbf{W}}$

Hyper-Parameter Tuning

- Hyper-parameters specify a detail of the learning algorithm.
- Bad hyper-parameters can cause the model to not converge.
- Hyper-parameters examples
 - Learning rate
 - Batch size
 - Choice of optimizer
 - Hidden layers
 - Number of Epochs
- Hyper-parameter tuning is often done through a grid search.



Too small: converge very slowly



Too big: overshoot and even diverge

Optimizers

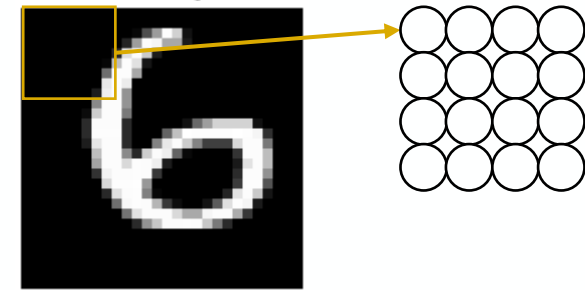
- An optimizer defines the update equation for the learnable parameters.
- Some examples of optimizers
 - Stochastic Gradient Descent
 - Adam
 - Adadelata
 - Adagrad
 - RMSProp

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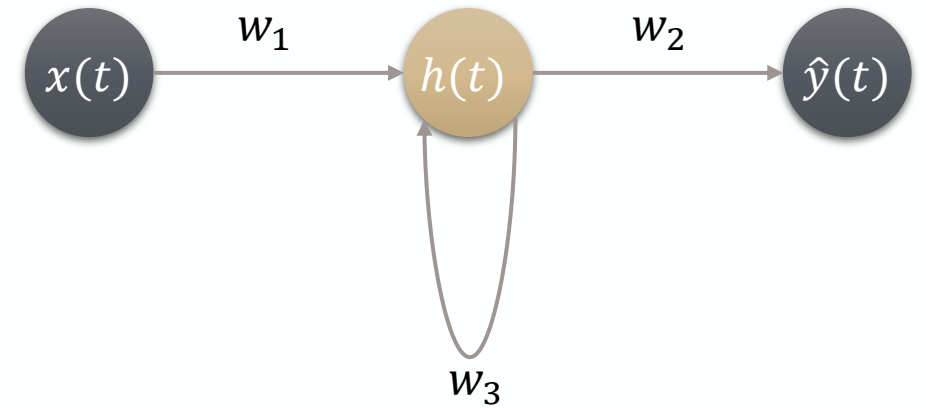
Convolutional Neural Networks

- Convolutional Neural Networks (CNNs) are spatially invariant.
- They are suitable for computer vision tasks.
- Each convolutional layer consists of number of filters with trainable weights.
- Each filter slides across the image to compute the output of the layer.
- Some examples of CNNs are LeNet, AlexNet, VGGNet, Inception.

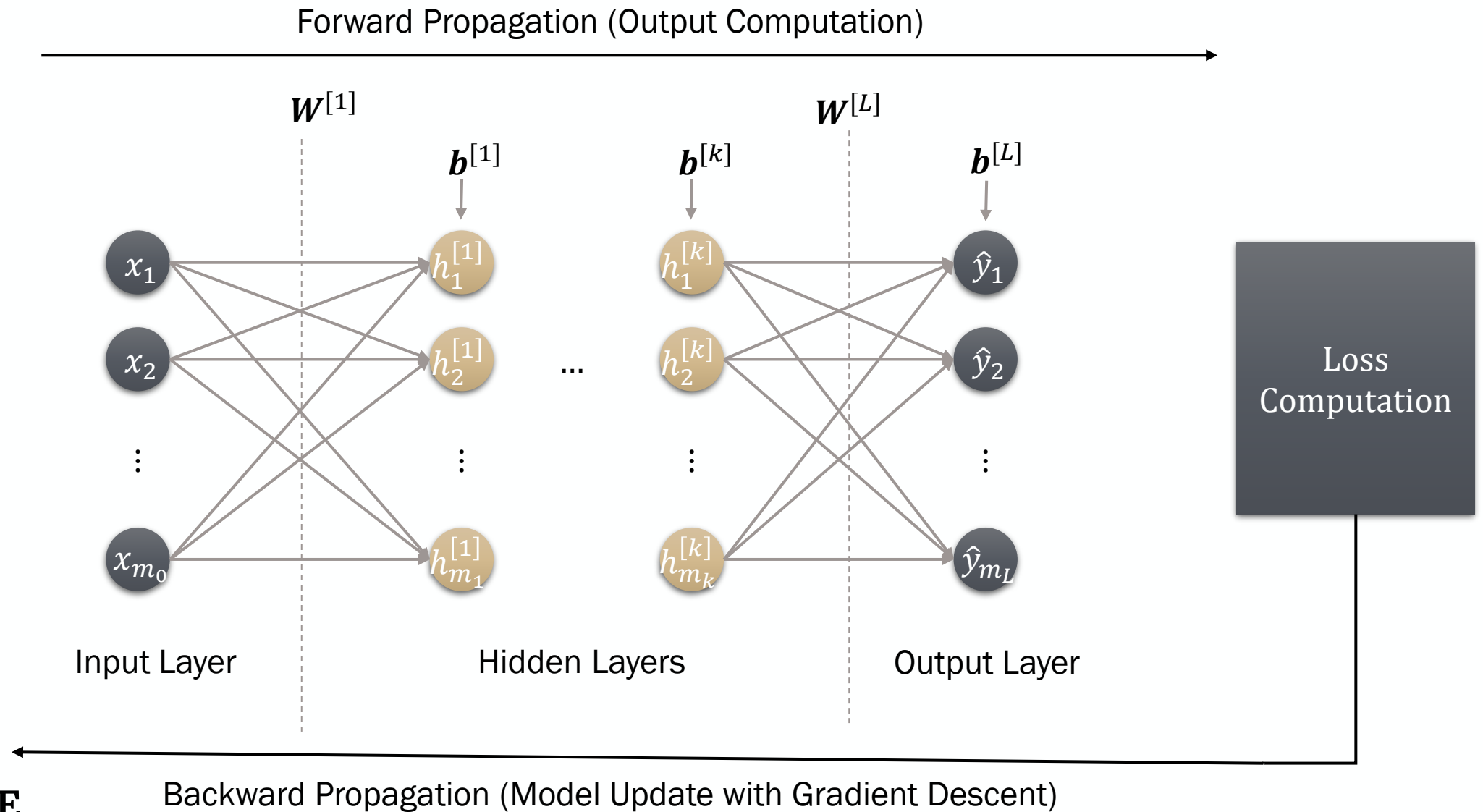


Sequence Models

- Sequence models are neural networks that input or output a sequence of data.
- They are suitable for timeseries data, text-streams, videos, and audio.
- The simplest sequence model is the Recurrent Neural Network (RNN).
- Variants of RNNs include Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRUs).
- An example of a sequence model is transformers.



Key Takeaway



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MNIST Dataset



Thank You

